

Evolving Knowledge Management: Artificial Intelligence and the Dynamics of Social Interactions

—XIAOMEI HE 

Faculté des Sciences Économiques et de Gestion, CNRS, INRAE, BETA, Université de Strasbourg, 67000 Strasbourg, France

—THIERRY BURGER-HELMCHEN 

Faculté des Sciences Économiques et de Gestion, CNRS, INRAE, BETA, Université de Strasbourg, 67000 Strasbourg, France

(Corresponding author: Thierry Burger-Helmchen.)

The review of this article was arranged by Department Editor Soila Lemmetty.

IEEE DOI 10.1109/EMR.2024.3519342

Abstract—This article examines the integration of artificial intelligence (AI) in knowledge management (KM), highlighting how AI reshapes traditional KM practices in organizations. It starts by reviewing historical KM frameworks, particularly the SECI model, which emphasizes the transformation between tacit and explicit knowledge. It differentiates generative AI from earlier AI models in terms of output nature, learning adaptability, and application scope. The study explores AI's potential to improve KM processes, focusing on how it influences social interactions, facilitates knowledge community development, and fosters collaborative intelligence by leveraging the strengths of both AI and human capabilities. AI's transformative role in KM includes enhancing knowledge creation, storage, and dissemination, accelerating processes like externalization and combination in the SECI model. However, the article notes AI's current limitations in managing tacit knowledge and human-centric decision-making. The research underscores AI's capacity to streamline KM processes, improve efficiency, and unlock new opportunities for organizations. By integrating AI tools with human expertise, organizations can enhance their ability to manage, share, and apply knowledge, setting the stage for innovative approaches to knowledge-driven growth and decision-making in diverse contexts.

Key words: Artificial intelligence (AI), human-machine interactions, knowledge management (KM), SECI model.

1. INTRODUCTION

THE importance of knowledge and information for economic prosperity has been widely acknowledged (Cooke and Leydesdorff, 2006). Knowledge management (KM) boosts organizational performance by efficiently leveraging intangible assets, such as intellectual capital and thereby enhancing its value. Now, artificial intelligence (AI) has emerged as a fundamental element of KM due to its enhancement of information gathering, development, and dissemination, along with its efficient implementation within organizations. Additionally, it facilitates the acquisition of novel insights that

enhance human knowledge (Cockburn et al., 2018; Sahay et al., 2021; Shollo and Galliers, 2016). Fosso Wamba and Queiroz (2023) highlight that “Knowledge” is the term most frequently encountered in AI research. Meanwhile, Sanzogni et al. (2017) advocate that AI and KM are intricately linked to the nature of knowledge.

KM can be addressed through two distinct approaches: leveraging personal (tacit) knowledge and utilizing codified (explicit) knowledge (Nonaka and Takeuchi, 1995; Sanchez, 2005). An analysis of the existing literature on KM reveals a concentrated effort toward developing information systems

(IS) within the KM domain during the 2000s. The main goal of equipping industries with efficient IS has been to effectively and efficiently store and retrieve knowledge (already processed and organized pieces of information). Early on knowledge has been distinguished from data and information, on the one hand and tacit knowledge, on the other hand. By combining these two types of knowledge, experts make decisions.

In recent years, with the rising of intellectual capital and its assessments, the utilization of practical knowledge associated with human capital has become the most important aspect of KM (Al-Jinini et al., 2019; Massingham and Massingham, 2014; Santoro et al., 2018). It was then recognized that computer-based solutions could yield highly accurate forecasts and recommendations by analyzing organizational knowledge. This realization paved the way for the integration of AI, data mining, business intelligence, and predictive analytics into the KM sphere, a trend that continues to this day. In fact, the technological advancements in machine learning (ML) and neural networks are increasingly becoming central to the mechanisms of knowledge dissemination and the capturing of expertise across various industrial sectors (Sohrabi et al., 2019).

KM research can be divided into four distinct phases as analyzed using scientometric methods (Serenko, 2013, 2021), which has been shown in Table 1: Phase I - Initiation (1996–2001): This phase established the foundational theories and practices of KM, with a focus on defining and exploring basic

concepts and frameworks. Phase II - Early Development (2002–2006): Research expanded rapidly, integrating empirical methods to validate initial theories. The role of technology and human factors in KM processes became a central focus, leading to more sophisticated models and tools. Phase III - Rigor and Consolidation (2007–2012): This period emphasized refining KM practices through rigorous empirical studies, consolidating previous research, and introducing advanced methodologies such as network analysis and meta-analysis. Phase IV - Methodological Advancement and New Technological Tools (2014–present): The current phase leverages more innovative and advanced approaches supported by cutting-edge technologies and has a high degree of specialization with KM studies conducted in a unique context.

The use of Big Data and social media during the current phase has created the ability to connect with previously inaccessible experts for collaboration on knowledge-driven projects. The vast amounts of knowledge generated both within and between organizations surpass the human capacity to effectively review and extract useful insights from these extensive data streams (Sohrabi et al., 2019). Consequently, there is a pressing need for new approaches to knowledge discovery and analysis, highlighting data science and advanced analytics as potential solutions.

Several studies have been conducted to explore the impact of AI technologies and tools on KM and how they affect contemporary organizations. A multifaceted

fundamental question concerning KM is the role of AI in organizations that have embraced KM practices. Recent studies show that AI can have multiple applications for KM, such as information retrieval (Guo et al., 2020;), topic modeling (Mustak et al., 2021), text mining (Antons et al., 2020; Kumar et al., 2021), personalization of training model in higher education (Xiao and Yi, 2021), and natural language processing (NLP) (Liu et al., 2021). For instance, Al-Emran et al. (2021) noted that implementing AI systems will contribute to personal KM through briefed documentation and automatic classification of documents and enhance distributed KM via tracking document sources, major revision to the document, and access records. To guide organizational KM that involves AI technologies, we, therefore, seek to study *how AI impacts KM activities and what is the nature of AI-human relationship in the context of KM*.

In attempting to answer this question, this study seeks to integrate the literature on AI with that on KM and focuses on examining the degree to which AI may enhance organizational KM practices and how AI augments human KM capabilities. This review undertakes a work of some tools and concepts of KM before the AI era and discusses how they evolve in this new technological setting. The article is organized as follows: Section II provides a literature review on the overview of KM concepts, SECI model, and knowledge communities. Section III then describes AI tools for KMS, KM dynamics, AI-human partnership, and the comparison between human learning and ML. Future research paths are discussed

Table 1. Four KM Research Phases by Serenko (2013, 2021).

	Phase I	Phase II	Phase III	Phase IV
Year	1996–2001	2002–2006	2007–2012	2014–present
Characteristics	Initiation	Early development	Rigor and consolidation	Methodological advancement and new technological tools

in section four and section five concludes.

2. LITERATURE REVIEW

A knowledge-based view of the organization emerges gradually in the 1980s–1990s in the strategic management literature. According to this trend, knowledge is the elementary brick of understanding and explaining the existence of organizations and their differential performance. If knowledge is at the very basis of the existence of companies, it also explains their competitive advantage. This advantage resides, from this perspective, in the possession of a stock of knowledge, but also in the ability to combine these resources to create value (Teece, 1998), that is, in the ability to put them into action (Pfeffer and Sutton, 2000), which quickly leads some authors to reason in terms of competencies. Traditional conception of knowledge focuses on the explicit nature of knowledge, while the contemporary view is based on the distinction of tacit and explicit knowledge (Eisenhardt et al., 2000). This section elucidates the foundational concepts of KM, including definitions for “knowledge” and “knowledge management,” clarifying knowledge typologies, outlining the knowledge transition model, and discussing the role of knowledge communities.

2.1 Knowledge Management: An Overview and Basic Concepts

The term “knowledge” has a wide range of interpretations. Historically, it has been linked to concepts such as data, information, intelligence, skill, experience, expertise, ideas, intuition, or insight, depending on the context (Gao et al., 2008). Plato defined knowledge as “justified true belief,” a notion that Nonaka and Takeuchi (1995) later expanded at the organizational level, describing it as “a dynamic human process of justifying personal belief toward the

truth.” From a broader perspective, Bell views knowledge as “a set of organized statements of facts or ideas, presenting a reasoned judgment or an experimental result, transmitted systematically through some communication medium.” Davenport and Prusak (1998) describe knowledge as “a fluid mix of framed experiences, values, contextual information, and expert insight.” Drucker (1999) considers knowledge to be information that “changes something or somebody, either by becoming grounds for action or by enabling an individual or institution to take more effective action,” also known as “specialized knowledge.” Knowledge can be further categorized as subjective or objective; or explicit or tacit/implicit. Polanyi (1961) was the first to distinguish between two dimensions of human knowledge: explicit knowledge and tacit knowledge. Explicit knowledge refers to information that is formalized and documented, such as data, scientific formulas, specifications, manuals, or textbooks. In contrast, tacit knowledge is unformulated and action-based, making it highly personal and difficult to transfer.

In the business context, both knowledge and competencies are recognized as strategic assets and key drivers of competitive advantage. However, the commercialization of new technology increasingly involves complex organizations, which requires an understanding of the nature of knowledge and competence. Webster (1981, p.63) defines competence as “the quality or state of being functionally adequate or of having sufficient knowledge, judgment, skill, or strength for a particular duty.” While knowledge assets are based on individuals’ experience and expertise, organizations provide the physical, social, and resource allocation structures to transform this knowledge into competencies.

These knowledge assets form the foundation of competencies, which, in turn, underpin the firm’s product and service offerings to the market (Teece, 2004).

In a business organization, KM is often described as a management tool that entails managing knowledge workers by facilitating, motivating, leading, and supporting them, as well as providing or nurturing a suitable working environment (Gao et al., 2008). The management of knowledge can be analyzed from two theoretical perspectives: individualistic (i.e., theories that emphasize the individual’s knowledge) and holistic (i.e., theories that prioritize the knowledge itself as the focal point) (Hollis, 1994).

The notion of KM was initially conceptualized in 1990s, driven by the globalization of economy and markets, the proliferation of knowledge-intensive products and services, and the rapid advancement of information technology (IT) (Alavi and Denford, 2012). One of the most frequently cited KM process frameworks is that proposed by Alavi and Leidner (2001), which delineates four core processes of KM: creation, storage retrieval, transference, and knowledge application. KM entails three main purposes within an organization:

- 1) clearly defining knowledge and asserting the value of it within a company;
- 2) fostering knowledge-intensive culture to promote KM activities, e.g., knowledge sharing;
- 3) developing a knowledge framework that includes technical sources and social mechanisms (Alavi and Leidner, 2001).

Tsai and Hung (2016) found four KM enablers that effectively support and promote the organizational KM: collaborative knowledge, specialization, employees’ IS knowledge, information system

(IS) infrastructure. Collaborative knowledge characterized by effective communication can provide support and encourage the integration and sharing of knowledge. Specialization allows the division of professional tasks and the shared decision-making authority in different sectors. Employee's IS knowledge helps them to extend their expertise across diverse domains and enhance KM activities. IT infrastructure connects employees through codifying and reusing valuable knowledge and can further facilitate the exploration of new knowledge.

KM systems (KMS) denote a class of IT-based systems developed to help companies handle information and manage knowledge effectively (Alavi and Leidner, 2001). The KMS was introduced to preserve the corporate history, as well as the experience and expertise of long-term employees (Lin and Huang, 2008). KMS empowers employees to generate and retain knowledge assets through diverse documents, enabling the structured organization and dissemination of information (Lin, 2013; Young, 2014). Santoro et al. (2018) underline that a KMS facilitates the creation of open and collaborative ecosystems. It enables the exploitation of internal and external flows of knowledge through the development of internal KM capacity, which, in turn, increases innovation capacity. With the progression of technological advancements, KMS are increasingly gravitating toward the adoption of sophisticated Information & Communication Technologies (ICT). In particular, numerous business initiatives are focusing on integrating AI as a pivotal tool within KMS.

2.2 Knowledge Typologies

Focusing on the most recurrent differentiation criteria in the literature, two fundamental dimensions are retained here:

organizational/individual—where knowledge resides—and tacit/codified—the form that knowledge takes for its diffusion (Polanyi, 1961). These dimensions allow for the integration of many other approaches, including Spender's (1996) based on differences between “objectified, collective, conscious, and automatic” knowledge. Blackler (2002) uses the terms “engraved in memory, embodied, encoded, and enculturated.” Finally, Johnson and Lundvall (1994) popularized the quartet “know what, know who, know how, know why.” In parallel with these typologies, other researchers have distinguished knowledge according to their scientific content, their geographical distribution, and their nature more or less easy to combine (Asheim et al., 2011). Fig. 1, proposed by Cohendet and Llerena (1999, p.220), offers an integrated view of these early typologies.

Using different terminologies, these various researchers study similar forms of knowledge.

- *Organizational/codified knowledge* corresponds to the set of scientific knowledge and

basic knowledge necessary to analyze a given situation. It is a group of knowledge that organizations must master in order to exist in their fields of activity. They form the “threshold” knowledge, which is rarely challenged but is always more precise with advances in science and evolving technologies.

- *Organizational/tacit knowledge* corresponds to the sum of routines and procedures used within the company. These routines depend on the company's culture and history and form the set of relational knowledge (know who). Thus, in case of difficulties in a company, thanks to this type of knowledge, employees know which steps to take or whom to contact within the organization to activate the appropriate action plan.
- *Individual/codified knowledge* corresponds to the individual knowledge acquired through education and other forms of learning. It also incorporates the mastery of the basic language related to professional activity.

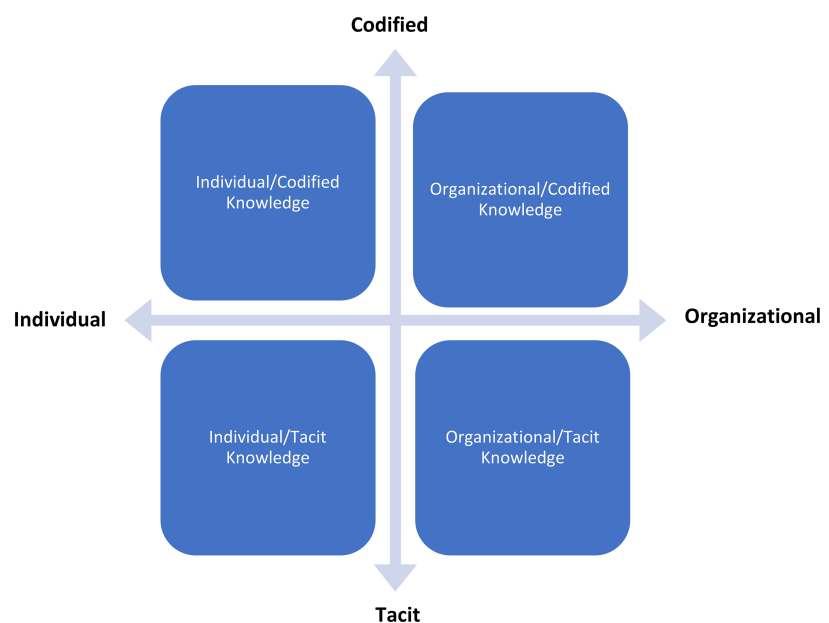


Figure 1. Typology of knowledge by Cohendet and Llerena (1999, p.220).

- *Individual/tacit knowledge* corresponds to all the Pavlovian reflexes related to the mastery of a profession, accumulated through experience. This category can be related to the notion of know-how.

2.3 Dynamics of Knowledge Transformation (SECI Model)

In 1995, Nonaka and Takeuchi introduced the dynamic process view of transformation knowledge in different forms based on the distinction between tacit and explicit knowledge. At the center of this theory of knowledge dynamics is a philosophy: knowledge develops, adapts, enriches, and creates itself through sharing and use. The basic assumption of Nonaka is that knowledge is created by a spiral consisting of four modes of conversion between tacit and explicit dimensions, between an individual and a group. These are

the foundations of the SECI model: Socialization—Externalization—Combination—Internalization (Fig. 2, from Nonaka and Konno, 1998, p.43). For Nonaka (as for Polanyi), explicit knowledge, which we are capable of transmitting through speech or writing, represents only a small part of what we know, and since we must codify it in language to express it, it is not only incomplete but also different from what we actually know. However, this language allows us to transmit common reference points (Cowan et al., 2000) like the codification of work by Taylor or ISO standards, for example. Tacit knowledge represents the complement, what we know but cannot properly transcribe in writing (or through digitization today) and which needs to be transmitted through example (learning, apprenticeship). Indeed, this tacit knowledge takes shape in practice, and it is by showing and trying that we manage to imbue ourselves with it.

The dynamics begin with the top left box—socialization. It represents the exchange of tacit knowledge between two individuals. This is the case, for example, when an apprentice imitates his master. It is through experience and repetition that an individual transmits his tacit knowledge to another. It is an act of socialization where two collaborators share their know-how. The second conversion mode—externalization—symbolizes a passage from tacit knowledge to explicit knowledge. Each individual who participates in a group holds a different tacit knowledge (at least in part). Externalization within a group and the subsequent formalization can already be a source of new knowledge for some of them. The third situation—combination—corresponds to the passage to a higher hierarchical level of the different explicit knowledge which then becomes the knowledge base of the organization after a new exercise of combination. It is

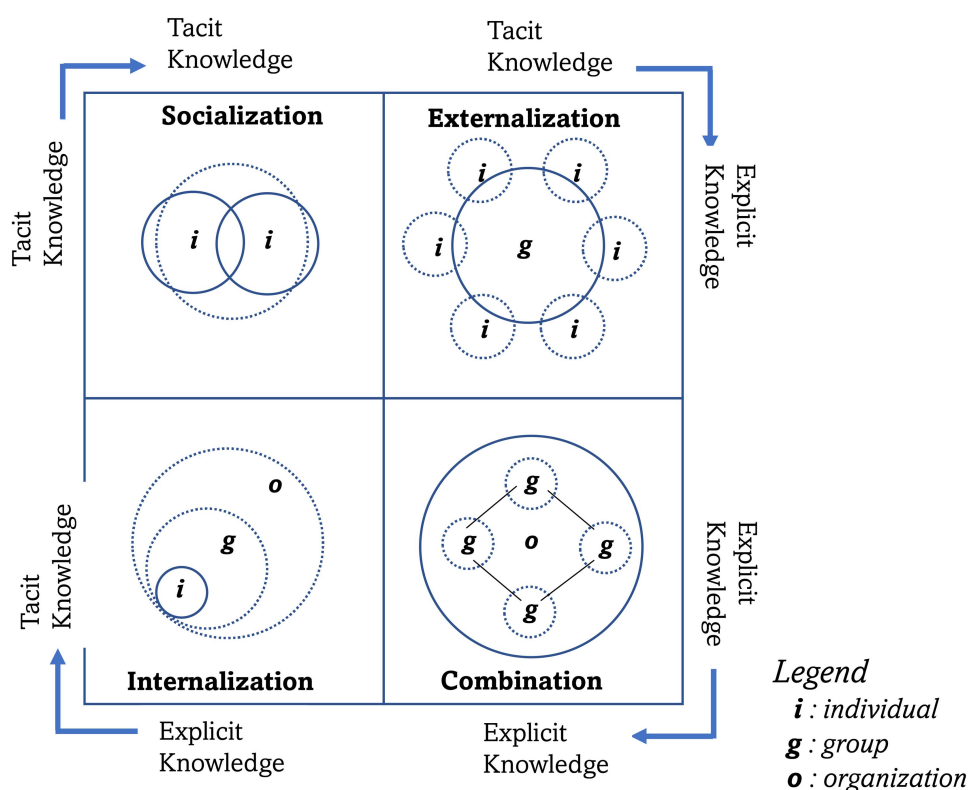


Figure 2. Dynamics of knowledge transformation by Nonaka and Konno (1998, p.43).

a mode of sustainable collective learning. Finally, the last phase corresponds to internalization, that is, the redistribution of knowledge at different levels. It is a phase during which explicit knowledge is transformed into tacit knowledge. During this transition, new and different practices take shape as each group or individual adapts collective knowledge to their own needs after selecting those that seem most relevant in the organization's knowledge base.

2.4 Ba and Knowledge

Communities Knowledge sometimes permeates an organization without its awareness, or at least, the hierarchical decision to impose the creation and adoption of knowledge is not without posing numerous practical problems. Continuing his reflection on the conditions of organizational knowledge creation, Nonaka gradually tempers the benefits of hierarchy to recognize the power of heterarchy in favoring certain modes of knowledge circulation. He goes further and invents the concept of Ba, as an "existential place where participants share contexts and create new meanings through their interactions" (Nonaka and Toyama, 2003, p.7). As a context for the emergence of knowledge sharing, this Ba is not a permanent place but an instant, a space during which worldviews are deliberately shared by actors, and therefore knowledge is created.

More generally, questions arise about the choice of structures that reduce the risks of forgetting, opportunism, or entrenchment due to privileged access to a type of information or knowledge, and/or those that catalyze integration and knowledge creation. Gradually, specific groups conducive to exchange are emerging: knowledge communities (Brown and Duguid, 1991; Goglio-Primard et al., 2017; Lave and Wenger, 1991). These

new organizational forms transcend hierarchical levels and companies have emerged to meet the needs of individuals and organizations to learn and share their knowledge. These are groups formed freely for the purpose of maintaining, sharing, or creating knowledge that group members need. These knowledge communities have a role in constructing meaning and memory and are not governed by a system of contract or financial incentive but through a system of trust and reputation.

Today, many forms of communities exist, with specific objectives and rules of operation, among which are communities of practice, users, epistemic, etc. (Amin and Roberts, 2008; Garrety et al., 2004). Lindkvist (2005) uses the term "collectivity of practice", which he describes as temporary groups within firms comprised of people who have not necessarily met before and "who have to engage in swift socialization and carry out a prespecified task within set limits as to time and costs" (p. 1190). This mode of organizing individuals has been developed based on following five forces:

- i) the ability to find solutions in relation to incongruous directives from management;
- ii) the construction of a collective memory to generate routines;
- iii) the support of novices;
- iv) the construction of typical conditions that allow individual work to be carried out;
- v) the ability to make work more enjoyable.

Cultivating such communities, therefore, calls on organizations to question the identity and socially constructed dimension of these communities and the knowledge they manipulate, and to understand their routines.

Those various foundational bricks of knowledge (data, information, intelligence) provide the framework for understanding how AI

technologies and KM will interact. The SECI model, describing the process of knowledge transformation, shows how AI can help the conversion of tacit knowledge into explicit knowledge and vice versa, which enhances the overall knowledge processes. Additionally, the concept of knowledge communities, where groups maintain, share, and create knowledge, highlights the importance of AI in improving communication and collaboration tools within these communities, as we will now see.

3. AI FOR KM

AI tools and systems are used extensively in vast modern business contexts, where KM has attracted growing interest in recent years. AI technologies can help organizations improve their KM strategies and achieve automation of KM operations (Coombs et al., 2020). A growing body of literature can be found in the fields of AI and KM respectively. Nowadays researchers focus on combining the two domains to improve the general applications of AI in KM, e.g., business, innovations, organizations, and healthcare (Botega and Da Silva, 2020; Chatterjee et al., 2020; Rhem, 2021). We will now examine how AI is revolutionizing traditional KM practices within organizations through four critical perspectives: AI-based KMS, AI's influence on KM dynamics, the Human-AI partnership, and a comparative analysis of human learning versus ML.

3.1 Knowledge Management System Based on AI

Most definitions of KMS come from the IS view, which considers KMS as a class of ISs using advanced IT to develop organizational KM applications. That is, they are software solutions to support and enhance the knowledge creation, storage or retrieval, transfer, and application in organizational processes (Alavi and Leidner, 2001). Three main applications of IT to organizational KM are frequently delineated in the literature

- (1) the systematic encoding and dissemination of best practices;
- (2) the development of corporate knowledge directories;
- (3) the establishment of knowledge networks.

An effective KMS typically consists of three key elements (Santoro et al., 2018).

- **IT Infrastructure:** This includes the tangible technological elements that facilitate efficient KM. Examples are hardware, software components, extranets, intranets, and Local Area Networks (Soto-Acosta and Meroño-Cerdan, 2008).
- **Collaborative Technologies:** These are tools that enable collaboration, such as discussion forums, shared databases, document repositories, and workflows (Meroño-Cerdan et al., 2007).
- **ICT Adoption:** This aspect involves integrating various collaborative technologies. The orientation of ICT usage focuses on three main implementation objectives: a) **Informative Orientation of ICT:** Aimed at delivering business information to various stakeholders across organizational and functional boundaries; b) **Communicative Orientation of ICT:** Facilitates cost reduction and enhances interaction with various business entities both within and outside the organization; c) **Workflow Orientation of ICT:** Establishes electronic processes within corporate technologies (Bafoutsou and Mentzas, 2002; Lopez-Nicolas and Soto-Acosta, 2010).

In the contemporary digital era, the efficacy of KM is contingent upon the deployment of advanced KMS. As categorized by Intezari and Gressel (2017), these sophisticated systems have five salient characteristics: social, cross-lingual, integrative,

dynamic and agile, as well as simple and understandable. In organizational KM, the vast amount of data leads to mismatches in information and reduced relevance of knowledge during the retrieval process. Big data is compelling firms to transform into data-driven organizations. To tackle the complexity of Big Data, more sophisticated technologies must be integrated to develop the next generation of KMS. AI tools, represented by ML, computer vision, NLP, and voice recognition methods, contribute to the processing of knowledge data and the improvement of information matching (Zhou et al., 2020). For instance, one example of the advanced KMS involves using the cognitive service approach based on ML algorithms to build the architecture of KMS by Tadejko (2020), which facilitates the discovery of insights by leveraging organizational knowledge, thus helps companies to enhance their competitiveness through knowledge derived from vast amounts of data. Likewise, Leng and Jiang (2016) proposed a deep learning model to facilitate knowledge transferring and sharing within the context of enterprise social interaction, therefore supporting the integration of the resources and capabilities among different enterprises. Lin (2022) introduced a bidirectional KMS with novel functionalities, such as improved hierarchical notetaking, AI-assisted brainstorming, and multidirectional relationships. This framework leverages relational databases and NLP-assisted annotation to record hierarchical information, which aids in the research and writing process, and enables the extraction of valuable knowledge from references or the discovery of insights from interconnected concepts.

3.2 AI for KM Dynamics Coming back to the SECI model of knowledge creation, the human contribution to the socialization aspect remains

crucial. Despite the increasing focus on explicit knowledge, such as digital technology and AI, the importance of enhancing tacit knowledge in the socialization process is more pronounced than ever. The potential of AI to reduce the time taken by humans for knowledge creation will contribute to the acceleration of the SECI process (Zhang et al., 2023). AI can substitute or even replace a human in some aspects of explicit knowledge transition, acting as an effective way of storing and organizing articulable knowledge. In contrast, AI is currently unable to take the place of humans in processing tacit knowledge or in making decisions and taking actions based on nonarticulable knowledge. However, the advent and rapid development of generative AI is prompting a reevaluation of this discussion. The distinction between generative AI and previous AI models for KM entails four main perspectives.

1. **Nature of Output:** Previous AI models focus on document and content management, whereas generative AI excels in content creation.
2. **Learning and Adaptability:** Traditional AI relies on rigid learning processes requiring extensive human intervention. In contrast, generative AI utilizes deep learning, which requires minimal human intervention and iterative adjustments.
3. **Interactivity:** Earlier AI systems provide preprogrammed responses or actions, while generative AI offers dynamic and context-aware interactions.
4. **Application Scope:** Conventional AI is applied to well-defined tasks, whereas generative AI is suited for creative and innovative applications.

AI for Socialization: AI typically facilitates the independence of time and space in this transformation, thereby speeding up the sharing of tacit knowledge and the transition

from tacit to explicit knowledge among various groups. AI ensures the efficiency and accuracy of these processes, enabling individuals, groups, and organizations to share their knowledge effectively with others. Especially when applied on a global scale, as in the case of social media, such AI algorithms also generate and sustain social tacit knowledge at a pace and with an impact that is relatively novel to contemporary society (Natek and Lesjak, 2021). For instance, ChatGPT can streamline the sharing of tacit knowledge during direct interactions by serving as a platform for virtual, geographically dispersed teams. This enables team members to communicate and share insights and information without the constraints of physical location (Korzynski et al., 2023). In the field of professional cooking, it is a common belief that the transmission of tacit knowledge about cooking from the chef or master to the apprentice occurs through hands-on experience and learning journey undertaken by the latter. In 2020, Sony AI launched the AI-powered Gastronomy Flagship Project, aiming at unleashing human creativity and imagination. The recipe creation is an iterative process and there are several tasks that can be facilitated or undertaken by AI: ingredients pairings, food conceptualization, context creation or storytelling, and prototyping. The use of AI could shrink those time-consuming processes and provide creative insights for humans.

AI for Externalization: AI ensures the efficiency and accuracy of processes, thereby enabling individuals or groups to effectively articulate, codify, publish, and disseminate their tacit knowledge. When implemented on a global scale, tools like wikis, thesauri, lexicons, and other published resources, particularly wikis, allow AI to develop and preserve explicit knowledge for society at large. This changes the landscape of global

knowledge accessibility (Natek and Lesjak, 2021). Many aspects of organizational decision-making are inherently considered as tacit and cannot be coded. However, Metcalf et al. (2019) suggest that tacit processes can be harnessed with the use of artificial swarm intelligence, a computational approach through which human tacit knowledge can be aggregated to make informed decisions.

AI for Combination: AI usually enhances the process of integrating various data and knowledge sources, as well as storing and distributing this information among a diverse range of individuals and groups. When deployed on a global scale, particularly through the use of algorithms, AI generates and upholds new explicit knowledge for both organizations and society at large. The uncovering of new patterns offers a novel perspective on social processes, thereby impacting the behavior of heterogeneous collectives (Natek and Lesjak, 2021). An exemplary instance is Google's Knowledge Graphs, which encapsulates the structural relations among various heterogeneous information sources, embodying knowledge about certain domains of discourse (Fensel et al., 2020). One critical research issue in knowledge graphs is Knowledge Representation Learning (KRL) which includes following four aspects:

- 1) representation space where the relations and entities are represented;
- 2) scoring function for measuring the plausibility of facts;
- 3) encoding models for modeling the semantic relations of facts;
- 4) auxiliary information by incorporating external information (Ji et al., 2021).

KRL is capable of combining multisource information of entities and relations into knowledge representations, such as entity

descriptions, hierarchical types, and textual relations (Tang et al., 2019).

AI for Internalization: AI's focus on the efficiency and accuracy of processes enables individuals or groups to access explicit knowledge during learning activities. For instance, an employee should be granted access to a range of academic resources, including internal study results, product development tests, cost analyses, market research, and external sources provided by the organization to some or all employees. When applied on a global scale, this approach transforms access to worldwide knowledge, particularly open-access knowledge, thereby reshaping educational systems and altering the learning practices of individuals, groups, organizations, and society as a whole (Natek and Lesjak, 2021).

3.3 AI for Knowledge

Communities As previously mentioned, AI systems are expected to increasingly integrate into the organizational KM. However, it is crucial to remember that the creation and management of knowledge are fundamentally centered around human involvement (Davenport and Prusak, 1998; Sanzogni et al., 2017). In KM, the most efficient roles designated for AI will primarily serve to complement human capability rather than substitute them. This approach points to hybrid or collaborative intelligence, where AI and humans amplify each other's distinct strengths through mutual learning (Paschen et al., 2020; Wilson and Daugherty, 2018). To better capture the human-AI relationship in KM, Jarrahi et al. (2023) articulated the concept of human-AI symbiosis (see Table 2).

An emerging genre of AI systems, known as personal intelligent assistants, is becoming increasingly significant in personal KM. Knowledge workers often face the

challenge of information overload in today's information-abundant environment (Gorman and Pauleen, 2016). Personal intelligent assistants have the potential to expand the cognitive capacity of these workers, altering how they assimilate relevant knowledge (Maedche et al., 2019). Personal intelligent assistants are adaptive, interactive systems capable of self-learning, and improving their interactions with users over time. Through this process, they can understand and adapt to each worker's evolving knowledge requirements, workflows, preferences, and feedback, offering tailored solutions (Ransbotham et al., 2019). Humans play a crucial role in the effectiveness of personal intelligent assistants. These systems rely heavily on feedback and interaction from users to function optimally. Essentially, personal assistants require ongoing training by humans to effectively exhibit complex and

nuanced human characteristics (Wilson and Daugherty, 2018).

AI represents specialized intelligence capable of sensing the environment, learning from experiences, and generating action possibilities in specific task scenarios. However, AI's function in KM and other areas is predominantly focused on particular tasks and struggles to adapt flexibly across different contexts. On the contrary, general intelligence remains a unique humane trait, which involves the knowledge of strategic thinking and decision-making. KM is characterized by judgment, deciding on the best course of action after weighing various options at hand (Jarrahi et al., 2023). Another attribute that distinguishes the general intelligence of humans from that of AI is self-reflection, the awareness of one's strengths and weaknesses (Jarrahi et al., 2019).

The codification strategy involves storing explicit knowledge in databases or repositories for reuse in new situations (Hansen et al., 2013). AI systems offer a wide range of possibilities to efficiently handle mundane, high-volume, routine tasks that are beyond the scope of human cognitive abilities alone. This allows knowledge workers to focus on more valuable and strategic analysis of explicit knowledge, which is more closely aligned with business issues. Since tacit knowledge transition is a predominantly human-centric process, humans will outperform AI in tackling complex and nonroutine processes.

AI systems evolve from traditional algorithms with human-created rules to generate standardized outputs to modern mechanisms with self-learning abilities to enhance their know-how and know-what (Jarrahi et al., 2023). The rationale of how AI algorithms negate results is often denoted as a black box. This underscores the critical role of humans in interpreting AI outputs to meet compliance requirements and provide essential "know-why" for AI-based inferences (Wilson and Daugherty, 2018). To enhance our understanding of machine's decision-making logics, humans must also develop AI literacy (Jarrahi et al., 2021), which describes the competence to accurately identify, use, and assess AI-related products while adhering to the ethical guidelines (Wang et al., 2023).

At the knowledge level, AI is associated with the assumption of perfect rationality, the capability to generate optimally effective actions based on predefined rulesets and formal procedures (Farrokhi et al., 2020; Newell, 1982; Nishant et al., 2024). Machines excel in rationality when dealing with contexts that can be easily codified. This underpins the efficiency of current-generation AI, particularly narrow AI, in tasks

Table 2. Human-AI Symbiosis Adapted From the Work of Jarrahi et al. (2023).

AI role	Human role
Personal Intelligent Assistants • Help with information overload • Increase cognitive bandwidth • Filter, sort, and navigate information sources	Personal KM • Train and individualize intelligent assistants • Scrutinize the performance of intelligent assistants • Develop AI literacy to understand machines' decision-making
Specialized Intelligence • Provide specialized intelligence for learning in bounded environments • Present task-centered intelligence (not easily transferable across different task contexts) • Manage knowledge content	Personal Intelligence • Apply knowledge for strategic-level thinking • Translate knowledge across contexts • Discern knowledge context via self-reflection
Codification of Knowledge • Streamline low-level, high-volume knowledge processes • Facilitate connecting people and generating know-how	Collaboration of Knowledge • Handle complex, nonroutine knowledge processes • Transfer tacit knowledge through social interactions
Know-How and Know-What • Discover unnoticed patterns in Big Data • Extend know-how outside of the current knowledge resources by developing own rules	Know-Why • Explain inferences and justify recommendations • Train budding experts and garner organizational support • Take responsibility
Noncognitive processes • Apply rationality based on well-defined rules	Cognitive processes • Be self-aware and reflect • Generate ideas by applying wisdom

that leverage logical statements embodying rationality. However, as data complexity grows and becomes infused with human interactions within specific contexts, the effectiveness of rationality diminishes (Nishant et al., 2024). In contrast, cognitive awareness is focal to the human self, encapsulating the entirety of “knowing” practices, particularly in how ideas are processed. Developing proper computational models that capture particularly the experience of “knowing that one knows” usually involves simplifying from multiple perspectives to a single viewpoint. However, the phenomenological attributes of experiences, which are intrinsically tied to feelings and emotions, defy such reduction, as they are fundamentally about the subjective experience itself. Therefore, the constraints of cognitive computational models do not seem to derive from the processing capabilities of IT or the shortcomings of software development. Instead, the core of these limitations lies in the flawed approach of system developers trying to replicate with computers, tasks that require self-awareness, wisdom, and human cognition, functions that can be performed by humans only (Sanzogni et al., 2017).

3.4 Synchronizing Machine and Human Learning in Organizations

As ML advances, the ability to learn and contribute to an organization's knowledge base is no longer exclusive to humans. ML differs significantly from Human learning. For example, human-based approaches to learning are predominantly a social activity (e.g., March, 1991), whereas ML primarily relies on processing data (e.g., Qiu et al., 2016). Organizations ought to synchronize human learning with ML to achieve effective learning collectively. Through a series of agent-based simulations, Sturm et al. (2021) identified three key insights.

- **Impact of ML on Human Learning:** ML can reduce the need for human exploratory learning, which is typically aimed at discovering new ideas and insights.
- **Human Adjustments to ML Systems:** Adjustments made by humans to ML systems are generally beneficial. However, this positive effect can diminish or even become counterproductive under certain conditions.
- **Reliance on ML in Turbulent Environments:** Depending on knowledge created by ML systems can be advantageous for organizational learning in unstable and changing environments. However, this requires substantial initial investments in setting up these systems and properly coordinating them with human efforts.

It is essential to continuously evaluate machine intelligence, its capabilities, areas requiring human intervention, and the potential for collaboration between humans and AI. For example, DeepMind's AlphaZero, using reinforcement learning, mastered chess by only learning the game rules, without prior exposure to common human strategies. It improved by playing against itself, guided solely by the rules and a reward mechanism. This approach to AI development suggests the potential for “machine tacit knowledge” that could surprise us and provide new learning avenues for humans (Jarrahi et al., 2022).

4. DISCUSSION: CONCERNS OVER AI/KM DEVELOPMENTAL PATHS

AI has the potential to enhance socially oriented conceptual KM tools by aiding human agents in handling various forms of knowledge, acting as an adjunct to human abilities. However, contemporary

advancements appear to be leaning toward a second direction of technological development—focusing on AI-driven creation of autonomous intelligent machines designed to replicate human behavior (Sanzogni et al., 2017). The evolution of AI and KM is influenced by organizational and technological progression trends (Collins, 1990). Two divergent paths for the advancements of AI/KM can be identified: AI's capability of emulating or competing with humans.

The first developmental path in AI and KM focuses on creating machines that enhance human actions and skills. In this model, humans handle tasks that demand wisdom and collective tacit knowledge embedded in a social milieu, while machines are designated for tasks involving explicit knowledge, and some weak tacit knowledge that is deliberately or nondeliberately hidden and somatic tacit knowledge about certain complex physical movements (Sanzogni et al., 2017). AI's emulation capacity (i.e., its ability to mimic human thought processes) offers both opportunities and challenges in management. On one side, it holds the potential to boost labor productivity, which has recently seen a declining growth rate, causing concern among economists and government officials (Byrne et al., 2016). However, AI's ability to replicate human decision-making has its limits, particularly in the area of innovation (Brem et al., 2023). For instance, AI is less effective in producing groundbreaking drug innovations, but performs adequately in identifying drug innovations with a moderate level of novelty (Irissarry and Burger-Helmchen, 2024). On the other hand, the ability of AI to emulate human behavior also brings about significant concerns, including security and ethical issues. Current cybersecurity measures, which differentiate between robot and human behaviors, may malfunction with AI's emulation skills,

enabling hackers to evade detection. Additionally, AI's emulation process can perpetuate human biases and errors.

In the second trajectory, there's a trend toward the codification and commodification of knowledge, where machines are seen as competitors to humans (Ferras-Hernandez et al., 2022). This approach treats knowledge as a tangible object, operating under the belief that all knowledge can be codified, thus turning it into a commodity. Consequently, technology designers create technologies aimed at either replacing human actions or transforming humans into extensions of AI/KMSs. While machine replacement of human tasks may lead to unemployment, the adaptation of humans as extensions of AI/KM could result in people altering their behavior to complement the capabilities of machines (Sanzogni et al., 2017). A similar perspective suggests that AI possesses the potential to outperform human capabilities, implying the achievement of a level of high-performance rational action that exceeds what humans are capable of (Russell and Norvig, 2010). Another issue raised in the research community pertains to the potential adversarial role of AI. Scholars have voiced concerns that AI could become an adversary when it undermines our capacity to learn and theorize, or when it misleads us with inaccurate, biased, or fabricated information (Kulkarni et al., 2023).

Another perspective for predicting the AI/KM developmental trajectory involves examining how the SECI model could be renovated by integrating humans and AI, and identifying which aspects are irreplaceable by AI agents. For example, Shen and Lin (2024) proposed a framework that transitions the SECI model from a human-to-human interaction paradigm

to a Human–AI loop format. This novel paradigm incorporates the capabilities of large language models and transactive memory systems, aiming to cultivate a community of practice facilitated by Human–AI knowledge cocreation agents (Bouschery et al., 2023). The study also highlights challenges related to participant motivation and the sustainability of the technical platform. Specifically, the willingness of members to share knowledge and the trust developed through AI assistance significantly influence the depth and breadth of shared knowledge. Future research should develop an action plan to address these challenges with insights from behavioral science and community development.

5. CONCLUSION

KM involves systematic processes and strategies for creating, utilizing, sharing, and overseeing an organization's information and knowledge assets. Traditional approaches to KM have faced limitations due to the finite capacity of humans to process and analyze extensive volumes of information. However, these gaps are increasingly being bridged by the widespread adoption of AI in KM and organizational learning. AI's capabilities to automate intricate processes, derive insights from data, and deliver precise predictions are revolutionizing how knowledge is managed and leveraged within organizations.

This article provides an overview of AI for KM practices and how human–AI relationship should be framed. The review mainly considers the KMS-integrated AI, AI's potential for KM dynamics characterized by the SECI model, the collaborative role of AI and human in supporting KM, and the inseparable nature of human and ML in the realm of KM. The integration of AI into the

KMS will help organizations solve the mismatches of information and improve knowledge processing efficiency. AI's ability to expedite knowledge creation accelerates the SECI process, potentially replacing human roles in transferring explicit knowledge efficiently. Yet, AI falls short in processing tacit knowledge and making decisions based on it. However, the rapid advancements in generative AI are leading to a reassessment of its capabilities in handling tacit knowledge and decision-making processes. AI systems are increasingly becoming integrated into organizational KM, emphasizing the importance of human involvement in the creation and management of knowledge. The most effective use of AI in KM is to augment rather than replace human capabilities, fostering a collaborative intelligence where AI and humans enhance each other's unique strengths. As ML evolves, it complements human capabilities in contributing to an organization's knowledge base, with ML focusing on data processing and human learning being more social. Discussions about the developmental trajectories of AI technologies for KM bring up various concerns. One perspective envisions AI as an assistant, replicating human cognitive abilities, while another suggests AI might surpass humans and become the primary decision-maker.

ACKNOWLEDGMENT

The work of X. He was financed by the China Scholarship Council grant to support the development of her PhD research. The work of T. Burger-Helmchen was supported in part by the ECO-INNOVATE ANR-24-CE26-7829 project managed by Sophie Bollinger (University of Strasbourg) and in part by Fondation de l'Université de Strasbourg through the research project AOC "Agilité Organisationnelle et Créativité."

REFERENCES

- [1] Alavi, M. and Denford, J.S. (2012), "Knowledge management: Process, practice, and web 2.0," *Handbook of Organizational Learning and Knowledge Management*, pp. 105–124.
- [2] Alavi, M. and Leidner, D.E. (2001), "Knowledge management and knowledge management systems: Conceptual foundations and research issues," *MIS Quarterly*, vol. 1, pp. 107–136.
- [3] Al-Emran, M., Shaalan, K., and Hassanien, A.E., Eds., (2021), *Recent Advances in Intelligent Systems and Smart Applications*, vol. 295. Berlin, Germany: Springer, doi: [10.1007/978-3-030-47411-9](https://doi.org/10.1007/978-3-030-47411-9).
- [4] Al-Jinini, D.K., Dahiyat, S.E., and Bontis, N. (2019), "Intellectual capital, entrepreneurial orientation, and technical innovation in small and medium-sized enterprises," *Knowledge and Process Management*, vol. 26, no. 2, pp. 69–85.
- [5] Amin, A. and Roberts, J. (2008), "Knowing in action: Beyond communities of practice," *Research Policy*, vol. 37, no. 2, pp. 353–369.
- [6] Antons, D., Grünwald, E., Cichy, P., and Salge, T.O. (2020), "The application of text mining methods in innovation research: Current state, evolution patterns, and development priorities," *R&D Management*, vol. 50, no. 3, pp. 329–351.
- [7] Asheim, B.T., Boschma, R., and Cooke, P. (2011), "Constructing regional advantage: Platform policies based on related variety and differentiated knowledge bases," *Regional Studies*, vol. 45, no. 7, pp. 893–904.
- [8] Bafoutsou, G. and Mentzas, G. (2002), "Review and functional classification of collaborative systems," *International Journal of Information Management*, vol. 22, no. 4, pp. 281–305.
- [9] Blackler, F. (2002), "Knowledge, knowledge work and organizations," in *The Strategic Management of Intellectual Capital and Organizational Knowledge*. London, U.K.: Oxford Univ. Press, pp. 47–62.
- [10] Botega, L.F.D.C. and Da Silva, J.C. (2020), "An artificial intelligence approach to support knowledge management on the selection of creativity and innovation techniques," *Journal of Knowledge Management*, vol. 24, no. 5, pp. 1107–1130, doi: [10.1108/JKM-10-2019-0559](https://doi.org/10.1108/JKM-10-2019-0559).
- [11] Bouschery, S.G., Blazevec, V., and Piller F.T. (2023), "Augmenting human innovation teams with artificial intelligence: Exploring transformer-based language models," *Journal of Product Innovation Management*, vol. 40, pp. 139–153.
- [12] Brem, A., Giones F., and Werle M. (Feb. 2023), "The AI digital revolution in innovation: A conceptual framework of artificial intelligence technologies for management of innovation," *IEEE Transaction on Engineering Management*, vol. 70, no. 2, pp. 770–776.
- [13] Brown, J.S. and Duguid, P. (1991), "Organizational learning and communities-of-practice: Toward a unified view of working, learning, and innovation," *Organization Science*, vol. 2, no. 1, pp. 40–57.
- [14] Byrne, D.M., Fernald, J.G., and Reinsdorf, M.B. (2016), "Does the United States have a productivity slowdown or a measurement problem?," *Brookings Papers on Economic Activity*, vol. 2016, no. 1, pp. 109–182.
- [15] Chatterjee, S., Ghosh, S.K., and Chaudhuri, R. (2020), "Knowledge management in improving business process: An interpretative framework for successful implementation of AI–CRM–KM system in organizations," *Business Process Management Journal*, vol. 26, no. 6, pp. 1261–1281.
- [16] Cockburn, I.M., Henderson, R., and Stern, S. (2018), "The impact of artificial intelligence on innovation: An exploratory analysis," in *The Economics of Artificial Intelligence: An Agenda*. Chicago, IL, USA: Univ. of Chicago Press, pp. 115–146.

- [17] Cohendet, P. and Llerena, P. (1999), "La conception de la firme comme processeur de connaissances," *Revue d'économie Industrielle*, vol. 88, no. 1, pp. 211–235, doi: [10.3406/rei.1999.1751](https://doi.org/10.3406/rei.1999.1751).
- [18] Collins, H.M. (1990), *Artificial Experts: Social Knowledge and Intelligent Machines*. Cambridge, MA, USA: MIT Press.
- [19] Cooke, P. and Leydesdorff, L. (2006), "Regional development in the knowledge-based economy: The construction of advantage," *The Journal of Technology Transfer*, vol. 31, pp. 5–15.
- [20] Coombs, C., Hislop, D., Taneva, S.K., and Barnard, S. (2020), "The strategic impacts of Intelligent Automation for knowledge and service work: An interdisciplinary review," *The Journal of Strategic Information Systems*, vol. 29, no. 4, Art. no. 101600.
- [21] Cowan, R., David, P.A., and Foray, D. (2000), "The explicit economics of knowledge codification and tacitness," *Industrial and Corporate Change*, vol. 9, no. 2, pp. 211–253.
- [22] Davenport, T.H. and Prusak, L. (1998), *Working Knowledge: How Organizations Manage What They Know*. Brighton, MA, USA: Harvard Business Press.
- [23] Drucker, P. (1999), *Managing for Results*. Evanston, IL, USA: Routledge, doi: [10.4324/9780080575315](https://doi.org/10.4324/9780080575315).
- [24] Eisenhardt, K.M., Santos, F.M., Pettigrew, I.A., Thomas, H., and Whittington, R. (2000), "Knowledge based view," in *Handbook of Strategy and Management*. Newbury Park, CA, USA: Sage.
- [25] Farrokhi, A., Shirazi, F., Hajli, N., and Tajvidi, M. (2020), "Using artificial intelligence to detect crisis related to events: Decision making in B2B by artificial intelligence," *Industrial Marketing Management*, vol. 91, pp. 257–273.
- [26] Fensel, D. et al., (2020), "Introduction: What is a knowledge graph?" in *Knowledge Graphs: Methodology, Tools and Selected Use Cases*. Berlin, Germany: Springer, pp. 1–10.
- [27] Ferras-Hernandez, X., Nylund P.A., and Brem, A. (thirdquarter, 2022), "Strategy follows the structure of artificial intelligence," *IEEE Engineering Management Review*, vol. 50, no. 3, pp. 17–19.
- [28] Fosso Wamba, S. and Queiroz, M.M. (2023), "Responsible artificial intelligence as a secret ingredient for digital health: Bibliometric analysis, insights, and research directions," *Information Systems Frontiers*, vol. 25, no. 6, pp. 2123–2138.
- [29] Gao, F., Li, M., and Clarke, S. (2008), "Knowledge, management, and knowledge management in business operations," *Journal of Knowledge Management*, vol. 12, no. 2, pp. 3–17, doi: [10.1108/13673270810859479](https://doi.org/10.1108/13673270810859479).
- [30] Garrety, K., Robertson, P.L., and Badham, R. (2004), "Integrating communities of practice in technology development projects," *International Journal of Project Management*, vol. 22, no. 5, pp. 351–358.
- [31] Goglio-Primard, K., Guittard, C., and Burger-Helmchen, T. (2017), "Introduction to the thematic file: Knowledge sharing in geographically dispersed communities," *Management International*, vol. 21, no. 3, pp. 12–17.
- [32] Gorman, G.E. and Pauleen, D.J. (2016), "The nature and value of personal knowledge management," in *Personal Knowledge Management*. Evanston, IL, USA: Routledge, pp. 1–16.
- [33] Gove, P.B. and G. & C. Merriam Company (1981), *Webster's Third New International Dictionary of the English Language, Unabridged*. Springfield, MA, USA: Merriam-Webster.
- [34] Guo, J. et al., (2020), "A deep look into neural ranking models for information retrieval," *Information Processing & Management*, vol. 57, no. 6, Art. no. 102067.
- [35] Hansen, M.T., Nohria, N., and Tierney, T. (2013), "What's your strategy for managing knowledge?" in *The Knowledge Management Yearbook 2000-2001*. Evanston, IL, USA: Routledge, pp. 55–69.
- [36] Hollis, M. (1994), *The Philosophy of Social Science: An Introduction*. Cambridge, U.K.: Cambridge Univ. Press.

- [37] Intezari, A. and Gressel, S. (2017), "Information and reformation in KM systems: Big Data and strategic decision-making," *Journal of Knowledge Management*, vol. 21, no. 1, pp. 71–91.
- [38] Jarrahi, M.H., Askay, D., Eshraghi, A., and Smith, P. (2023), "Artificial intelligence and knowledge management: A partnership between human and AI," *Business Horizons*, vol. 66, no. 1, pp. 87–99, doi: [10.1016/j.bushor.2022.03.002](https://doi.org/10.1016/j.bushor.2022.03.002).
- [39] Jarrahi, M.H., Lutz, C., and Newlands, G. (2022), "Artificial intelligence, human intelligence and hybrid intelligence based on mutual augmentation," *Big Data & Society*, vol. 9, no. 2, Art. no. 20539517221142824.
- [40] Jarrahi, M.H., Newlands, G., Lee, M.K., Wolf, C.T., Kinder, E., and Sutherland, W. (2021), "Algorithmic management in a work context," *Big Data & Society*, vol. 8, no. 2, Art. no. 20539517211020332.
- [41] Jarrahi, M.H., Philips, G., Sutherland, W., Sawyer, S., and Erickson, I. (2019), "Personalization of knowledge, personal knowledge ecology, and digital nomadism," *Journal of the Association for Information Science and Technology*, vol. 70, no. 4, pp. 313–324.
- [42] Ji, S., Pan, S., Cambria, E., Marttinen, P., and Philip, S.Y. (Feb. 2022), "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 2, pp. 494–514.
- [43] Johnson, B. and Lundvall, B.A. (1994), "The learning economy," *Journal of Industry Studies*, vol. 1, no. 2, pp. 23–42.
- [44] Korzynski, P. et al., (2023), "Generative artificial intelligence as a new context for management theories: Analysis of ChatGPT," *Central European Management Journal*, vol. 31, no. 1, pp. 3–13.
- [45] Kulkarni, M. et al., (2023), "The future of research in an artificial intelligence-driven world," *Journal of Management Inquiry*, vol. 33, pp. 207–229.
- [46] Kumar, S., Kar, A.K., and Ilavarasan, P.V. (2021), "Applications of text mining in services management: A systematic literature review," *International Journal of Information Management Data Insights*, vol. 1, no. 1, Art. no. 100008.
- [47] Lave, J. and Wenger, E. (1991), *Situated Learning: Legitimate Peripheral Participation*. Cambridge, U.K.: Cambridge Univ. Press.
- [48] Leng, J. and Jiang, P. (2016), "A deep learning approach for relationship extraction from interaction context in social manufacturing paradigm," *Knowledge-Based Systems*, vol. 100, pp. 188–199.
- [49] Lin, B. (2022), "Knowledge management system with nlp-assisted annotations: A brief survey and outlook," *arXiv:2206.07304*.
- [50] Lin, H.-F. (2013), "Examining the factors influencing knowledge management system adoption and continuance intention," *Knowledge Management Research & Practice*, vol. 11, pp. 389–404.
- [51] Lin, T.-C. and Huang, C.-C. (2008), "Understanding knowledge management system usage antecedents: An integration of social cognitive theory and task technology fit," *Information & Management*, vol. 45, no. 6, pp. 410–417, doi: [10.1016/j.im.2008.06.004](https://doi.org/10.1016/j.im.2008.06.004).
- [52] Lindkvist, L. (2005), "Knowledge communities and knowledge collectivities: A typology of knowledge work in groups," *Journal of Management Studies*, vol. 42, no. 6, pp. 1189–1210, doi: [10.1111/j.1467-6486.2005.00538.x](https://doi.org/10.1111/j.1467-6486.2005.00538.x).
- [53] Liu, X., Shin, H., and Burns, A.C. (2021), "Examining the impact of luxury brand's social media marketing on customer engagement: Using Big Data analytics and natural language processing," *Journal of Business Research*, vol. 125, pp. 815–826.
- [54] Lopez-Nicolas, C. and Soto-Acosta, P. (2010), "Analyzing ICT adoption and use effects on knowledge creation: An empirical investigation in SMEs," *International Journal of Information Management*, vol. 30, no. 6, pp. 521–528.
- [55] Irissarry, C. and Burger-Helmchen, T. (2024), "Using artificial intelligence to advance the research and development of orphan drugs," *Businesses*, vol. 4, pp. 453–472, doi: [10.3390/businesses4030028](https://doi.org/10.3390/businesses4030028).

- [56] Maedche, A. et al., (2019), "AI-based digital assistants: Opportunities, threats, and research perspectives," *Business & Information Systems Engineering*, vol. 61, pp. 535–544.
- [57] March, J.G. (1991), "Exploration and exploitation in organizational learning," *Organization Science*, vol. 2, no. 1, pp. 71–87.
- [58] Massingham, P.R. and Massingham, R.K. (2014), "Does knowledge management produce practical outcomes?" *Journal of Knowledge Management*, vol. 18, no. 2, pp. 221–254, doi: [10.1108/JKM-10-2013-0390](https://doi.org/10.1108/JKM-10-2013-0390).
- [59] Meroño-Cerdan, A.L., Soto-Acosta, P., and López-Nicolás, C. (2007), "Analyzing collaborative technologies' effect on performance through intranet use orientations," *Journal of Enterprise Information Management*, vol. 21, no. 1, pp. 39–51.
- [60] Metcalf, L., Askay, D.A., and Rosenberg, L.B. (2019), "Keeping humans in the loop: Pooling knowledge through artificial swarm intelligence to improve business decision making," *California Management Review*, vol. 61, no. 4, pp. 84–109.
- [61] Mustak, M., Salminen, J., Plé, L., and Wirtz, J. (2021), "Artificial intelligence in marketing: Topic modeling, scientometric analysis, and research agenda," *Journal of Business Research*, vol. 124, pp. 389–404.
- [62] Natek, S. and Lesjak, D. (2021), "Knowledge management systems and tacit knowledge," *International Journal of Innovation and Learning*, vol. 29, no. 2, Art. no. 166, doi: [10.1504/IJIL.2021.112994](https://doi.org/10.1504/IJIL.2021.112994).
- [63] Newell, A. (1982), "The knowledge level," *Artificial Intelligence*, vol. 18, no. 1, pp. 87–127.
- [64] Nishant, R., Schneckenberg, D., and Ravishankar, M.N. (2024), "The formal rationality of artificial intelligence-based algorithms and the problem of bias," *Journal of Information Technology*, vol. 39, no. 1, pp. 19–40.
- [65] Nonaka, I. and Konno, N. (1998), "The concept of 'Ba': Building a foundation for knowledge creation," *California Management Review*, vol. 40, no. 3, pp. 40–54.
- [66] Nonaka, I. and Takeuchi, H. (1995), *The Knowledge-Creating Company*. London, U.K.: Oxford Univ. Press.
- [67] Nonaka, I. and Toyama, R. (2003), "The knowledge-creating theory revisited: Knowledge creation as a synthesizing process," *Knowledge Management Research & Practice*, vol. 1, no. 1, pp. 2–10.
- [68] Paschen, J., Wilson, M., and Ferreira, J.J. (2020), "Collaborative intelligence: How human and artificial intelligence create value along the B2B sales funnel," *Business Horizons*, vol. 63, no. 3, pp. 403–414.
- [69] Pfeffer, J. and Sutton, R.I. (2000), *The Knowing-Doing Gap: How Smart Companies Turn Knowledge into Action*. Brighton, MA, USA: Harvard Business Press.
- [70] Polanyi, M. (1961), "The tacit dimension," in *Knowledge in Organisations*. Evanston, IL, USA: Routledge, pp. 135–146.
- [71] Qiu, J., Wu, Q., Ding, G., Xu, Y., and Feng, S. (2016), "A survey of machine learning for Big Data processing," *EURASIP Journal on Advances in Signal Processing*, vol. 2016, pp. 1–16.
- [72] Ransbotham, S., Khodabandeh, S., Fehling, R., LaFountain, B., and Kiron, D. (2019), "Winning with AI," *MIT Sloan Management Review*.
- [73] Rhem, A.J. (2021), "AI ethics and its impact on knowledge management," *AI and Ethics*, vol. 1, no. 1, pp. 33–37.
- [74] Russell, S.J. and Norvig, P. (2010), *Artificial Intelligence a Modern Approach*.
- [75] Sahay, S.K., Goel, N., Jadliwala, M. and Upadhyaya, S. (2021), "Advances in secure knowledge management in the artificial intelligence era," *Information Systems Frontiers*, vol. 23, pp. 807–810.
- [76] Sanchez, R. (2005), "'Tacit knowledge' versus 'explicit knowledge' approaches to knowledge management practice," in Rooney, D., Hearn, G., and Ninan, A., Eds., *Handbook on the Knowledge Economy*. Northampton, MA, USA: Edward Elgar Publishing, pp. 191–204, doi: [10.4337/9781845426842.00025](https://doi.org/10.4337/9781845426842.00025).

- [77] Santoro, G., Vrontis, D., Thrassou, A., and Dezi, L. (2018), "The Internet of Things: Building a knowledge management system for open innovation and knowledge management capacity," *Technological Forecasting and Social Change*, vol. 136, pp. 347–354.
- [78] Sanzogni, L., Guzman, G., and Busch, P. (2017), "Artificial intelligence and knowledge management: Questioning the tacit dimension," *Prometheus*, vol. 35, no. 1, pp. 37–56, doi: [10.1080/08109028.2017.1364547](https://doi.org/10.1080/08109028.2017.1364547).
- [79] Serenko, A. (2013), "Meta-analysis of scientometric research of knowledge management: Discovering the identity of the discipline," *Journal of Knowledge Management*, vol. 17, no. 5, pp. 773–812.
- [80] Serenko, A. (2021), "A structured literature review of scientometric research of the knowledge management discipline: A 2021 update," *Journal of Knowledge Management*, vol. 25, no. 8, pp. 1889–1925, doi: [10.1108/JKM-09-2020-0730](https://doi.org/10.1108/JKM-09-2020-0730).
- [81] Shen, W.-C. and Lin, F.-R. (2024), "The design of AI-enabled experience-based knowledge management system to facilitate knowing and doing in communities of practice," in Uden, L. and Ting, I.-H., Eds., *Knowledge Management in Organisations*. Berlin, Germany: Springer, pp. 292–303, doi: [10.1007/978-3-031-63269-3_22](https://doi.org/10.1007/978-3-031-63269-3_22).
- [82] Shollo, A. and Galliers, R.D. (2016), "Towards an understanding of the role of business intelligence systems in organisational knowing," *Information Systems Journal*, vol. 26, no. 4, pp. 339–367.
- [83] Sohrabi, B., Vanani, I.R., Jalali, S.M.J., and Abedin, E. (2019), "Evaluation of research trends in knowledge management: A hybrid analysis through burst detection and text clustering," *Journal of Information & Knowledge Management*, vol. 18, no. 04, Art. no. 1950043, doi: [10.1142/S0219649219500436](https://doi.org/10.1142/S0219649219500436).
- [84] Soto-Acosta, P. and Meroño-Cerdan, A.L. (2008), "Analyzing e-business value creation from a resource-based perspective," *International Journal of Information Management*, vol. 28, no. 1, pp. 49–60.
- [85] Spender, J.-C. (1996), "Organizational knowledge, learning and memory: Three concepts in search of a theory," *Journal of Organizational Change Management*, vol. 9, no. 1, pp. 63–78.
- [86] Sturm, T. et al., (2021), "Coordinating human and machine learning for effective organizational learning," *MIS Quarterly*, vol. 45, no. 3, pp. 1581–1602.
- [87] Tadejko, P. (2020), "Cloud cognitive services based on machine learning methods in architecture of modern knowledge management solutions," in *Data-Centric Business and Applications: Towards Software Development*, vol. 4. Berlin, Germany: Springer, pp. 169–190.
- [88] Tang, X., Chen, L., Cui, J., and Wei, B. (2019), "Knowledge representation learning with entity descriptions, hierarchical types, and textual relations," *Information Processing & Management*, vol. 56, no. 3, pp. 809–822.
- [89] Teece, D.J. (1998), "Capturing value from knowledge assets: The new economy, markets for know-how, and intangible assets," *California Management Review*, vol. 40, no. 3, pp. 55–79.
- [90] Teece, D.J. (2004), "Knowledge and Competence as Strategic Assets," in Holsapple, C.W., Ed., *Handbook on Knowledge Management 1: Knowledge Matters*. Berlin, Germany: Springer, pp. 129–152, doi: [10.1007/978-3-540-24746-3_7](https://doi.org/10.1007/978-3-540-24746-3_7).
- [91] Tsai, J.C.-A. and Hung, S.-Y. (2016), "Determinants of knowledge management system adoption in health care," *Journal of Organizational Computing and Electronic Commerce*, vol. 26, no. 3, pp. 244–266, doi: [10.1080/10919392.2016.1194062](https://doi.org/10.1080/10919392.2016.1194062).
- [92] Wang, B., Rau, P.-L.P., and Yuan, T. (2023), "Measuring user competence in using artificial intelligence: Validity and reliability of artificial intelligence literacy scale," *Behaviour & Information Technology*, vol. 42, no. 9, pp. 1324–1337.
- [93] Wilson, H.J. and Daugherty, P.R. (2018), "Collaborative intelligence: Humans and AI are joining forces," *Harvard Business Review*, vol. 96, no. 4, pp. 114–123.

- [94] Xiao, M. and Yi, H. (2021), "Building an efficient artificial intelligence model for personalized training in colleges and universities," *Computer Applications in Engineering Education*, vol. 29, no. 2, pp. 350–358.
- [95] Young, M.-L. (2014), "The formation of concern for face and its impact on knowledge sharing intention in knowledge management systems," *Knowledge Management Research & Practice*, vol. 12, pp. 36–47.
- [96] Zhang, W., Zhao, X., Zhao, L., Yin, D., Yang, G.H., and Beutel, A. (2020), "Deep reinforcement learning for information retrieval: Fundamentals and advances," in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 2468–2471.
- [97] Zhang, X., Liu, Z., Cheng, Y., Wang, X., and Wang, Z. (2023), "The impact of AIGC on organizational knowledge creation: From the perspective of adaptive structuration theory," in *Proc. 2023 IEEE International Conference on Data Mining Workshops*, pp. 1469–1476.
- [98] Zhou, Z., Liu, Y., Yu, H., and Ren, L. (2020), "The influence of machine learning-based knowledge management model on enterprise organizational capability innovation and industrial development," *PLoS One*, vol. 15, no. 12, Art. no. e0242253.

Xiaomei He received the Bachelor's degree in international economics and management and a Master's degree in economics and management of innovation from the Faculty of economics and Management, University of Strasbourg, Strasbourg, France, in 2021 and 2023, respectively. She is currently working toward the Ph.D. degree in innovation management with the Research Lab BETA-CNRS. Her research interests include on technology-driven innovation, knowledge management, new product development, and the applications of artificial intelligence in innovation processes.

Thierry Burger-Helmchen is a Professor with the Faculty of Economics and Management, University of Strasbourg, specializing in innovation management. His research at the BETA lab examines how organizations craft innovation strategies, focusing on emerging technologies like AI. He teaches at MBA, DBA, and PhD levels and regularly delivers conferences for public and private organizations.